

Department of Computer Science

AMU Aligarh

M.C.A. IIIrd Semester Lab First Sessional Test (CAMS – 3P01)

Note: Attempt all questions.

Set-A

Duration: 45 Minutes

Q1. Write a Python program that defines `is_special_number(n)` to check if a number is divisible by 5, a palindrome, and has even digit sum. Between 100 and 999, print all special numbers and their count.

OR

Q1'. Write a Python program that input a string and truncate it from a given index. If remaining has more vowels than consonants, replace vowels with ASCII in original; else, reverse and print the string.

Q3. Write a Python program to display all prime numbers within a given range from user input.

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Set-B

Duration: 45 Minutes

Q1. Write a Python program that input an integer, print a reverse triangular pattern, compute and store factorials of each row's start, then display the total, maximum, and minimum from the factorial list.

OR

Q1'. Write a Python program that accepts a tuple of integers, count occurrences of the maximum value, create a new tuple excluding the minimum, and check if its length is prime, printing appropriate results.

Q2. Write a Python program to check whether the given number is a palindrome or not from user input.

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Set-C

Duration: 45 Minutes

Q1. Write a Python program that accept two lists of 10 integers (odd, even), merge them alternately, square every third element, then compute and display the sum of all prime numbers in the final list.

OR

Q1'. Write a python program to create a 4×4 NumPy array (1–100), extract rows with ≥ 2 even numbers, compute each row's sum and average of extracted rows. You are also supposed to form a transpose matrix of the original matrix.

Q2. Write a Python program to check whether the given number is a Armstrong or not.